

Merlin Plus: A Large-Scale, Multi-Cancer, Longitudinal, Image-Mask-Report Dataset

Pedro R. A. S. Bassi^{1,2,3†}, Wenxuan Li^{1,2,3†}, Szymon Plotka^{4†},
Ruby Honjol⁵, Jakub Prądo⁶, Xinze Zhou¹, Kang Wang⁷, Yang Yang⁷, Malte Jensen⁵,
Akshay S. Chaudhari⁵, Curtis P. Langlotz⁵, Alan L. Yuille¹, and Zongwei Zhou^{1,8*}

¹Johns Hopkins University ²Harvard Medical School ³Massachusetts General Hospital ⁴Jagiellonian University
⁵Stanford University ⁶Warmian-Masurian Cancer Center ⁷University of California, San Francisco ⁸Johns Hopkins Medicine

zzhou82@jh.edu; Data (QR code): github.com/mrgiovanni/MerlinPlus



A Public Dataset of Longitudinal CT Scans, Radiology Reports, and Masks for 9 Cancer Types

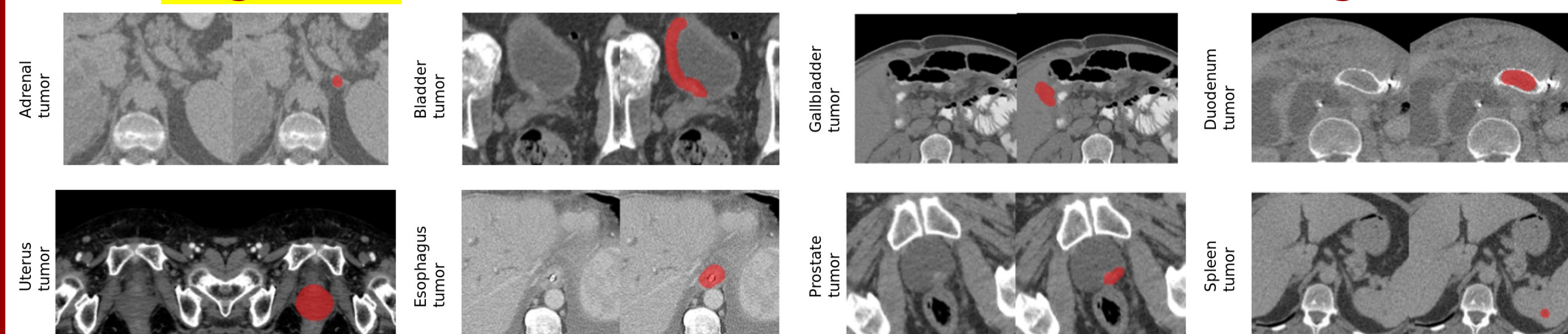
Background: Tumor segmentation models can advance cancer detection and surpass VLMs and classifiers, but public datasets lack masks for most tumor types in CT. Furthermore, longitudinal AI models are expected to learn cancer progression, but most public CT datasets lack multiple scans per patient.

Contribution: In Merlin Plus, radiologists created tumor masks for tumors in 9 organs in the Merlin dataset (Stanford hospital, 25,494 CT scans), the largest pelvic and abdominal public CT dataset. Additionally, we release anonymized patient IDs and scan dates, making Merlin Plus a longitudinal dataset: 3,830 patients have two or more CT scans.

Method: A new active learning method, based on radiology reports and Report Supervision (R-Super), allowed 4 radiologists to create masks 3 to 5 times faster.

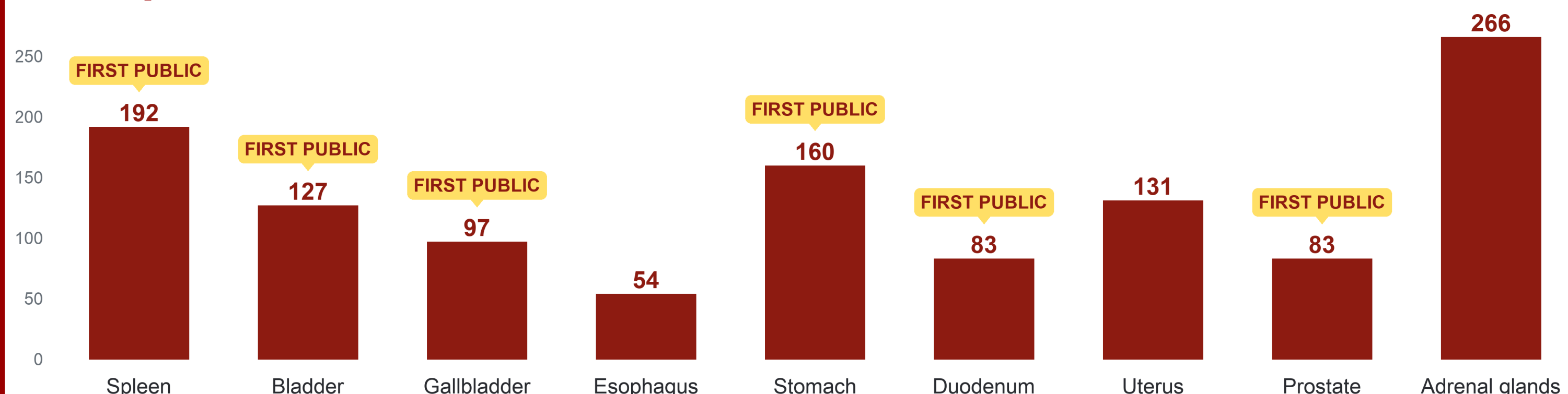
Results: Segmentation models trained on Merlin Plus substantially surpass state-of-the-art public AI models (ULS and FLARE) in the detection and segmentation of 9 tumor types, with +23% F1-Score and +29% DSC, showing that Merlin Plus fills a substantial gap in public data. Segmentation models trained on Merlin Plus also substantially surpass the Merlin model and classification models trained on Merlin in tumor detection, by +10% F1-Score, revealing the importance of tumor segmentation masks.

Large-Scale: 25,494 CT Scans, Masks for Tumors in 9 Organs



❖ 1,193 masks, including all confirmed malignant tumors

New Tumor Types: First Public Masks for Tumors in the Spleen, Bladder, Gallbladder, Stomach, Duodenum and Prostate



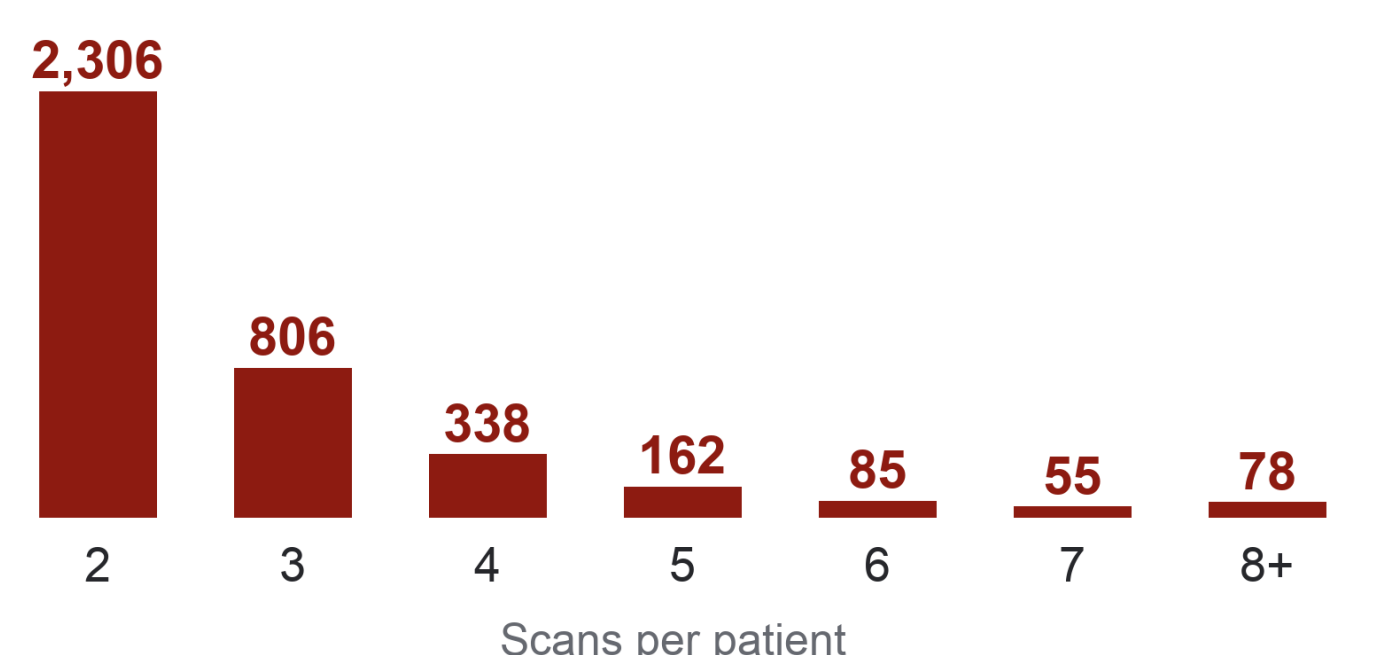
Longitudinal: Patients with Multiple Time Points

3,830
patients with ≥ 2 scans
21.2% of all patients

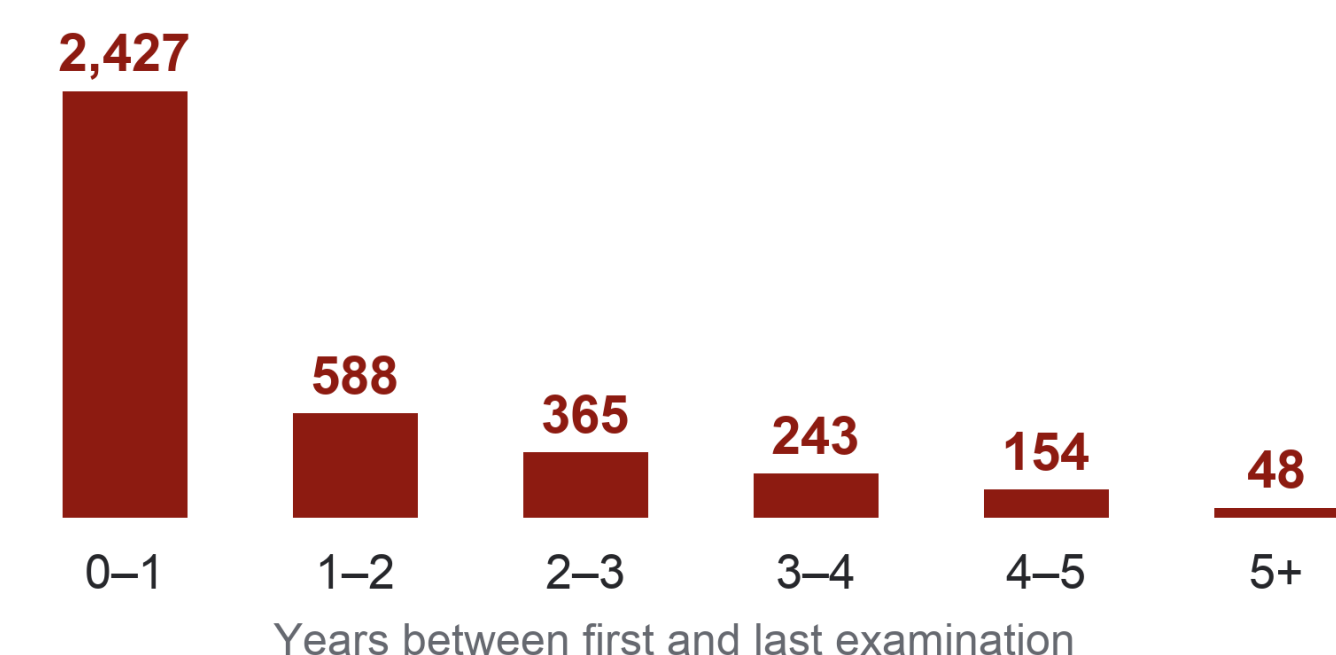
1,524
patients with ≥ 3 scans
Repeated observations over time

10,891
scans in longitudinal patients
Up to 18 scans per patient

How many scans per patient?
Patients with repeated scans (n = 3,830)



How long do observations span?
First-to-last examination interval

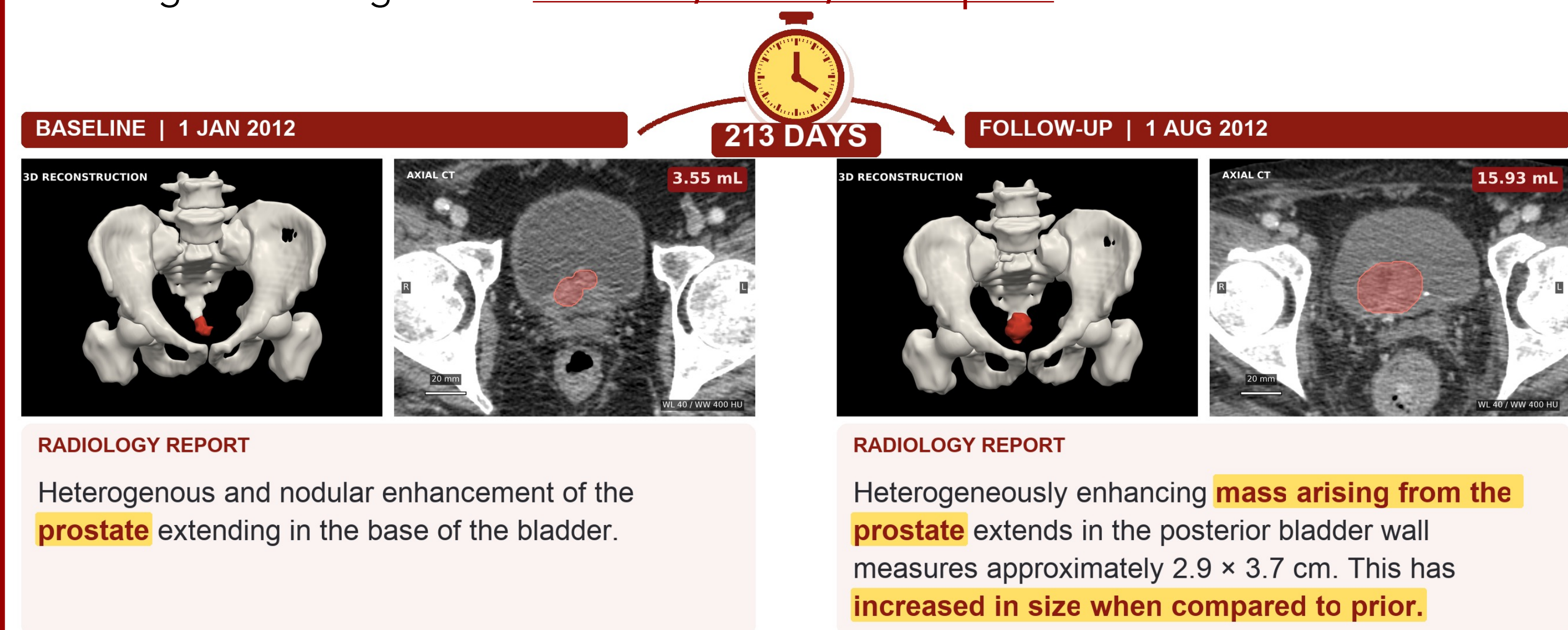


193 days
median observation span

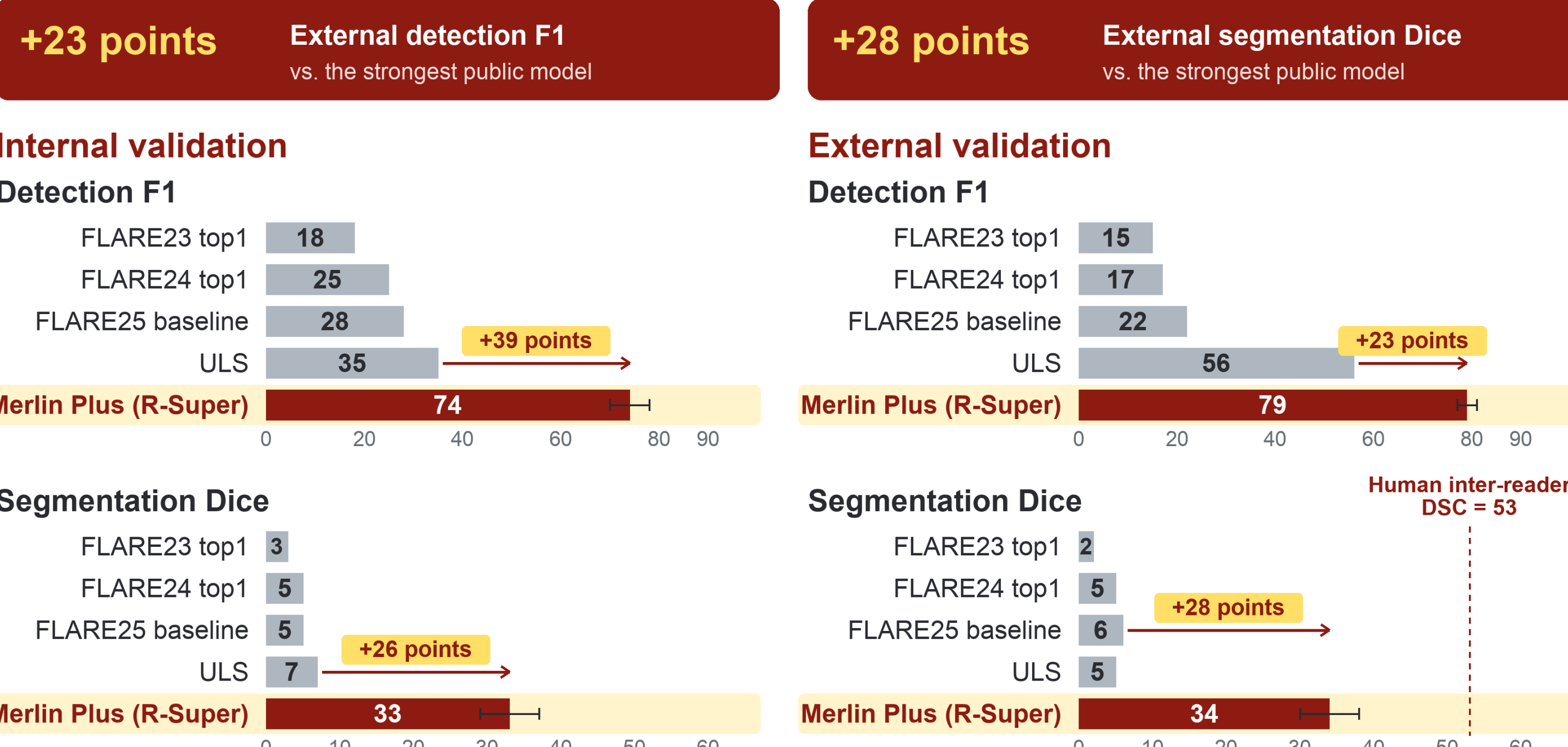
5.6 years
longest span (2,051 days)

1,398 patients
with ≥ 1 year between scans

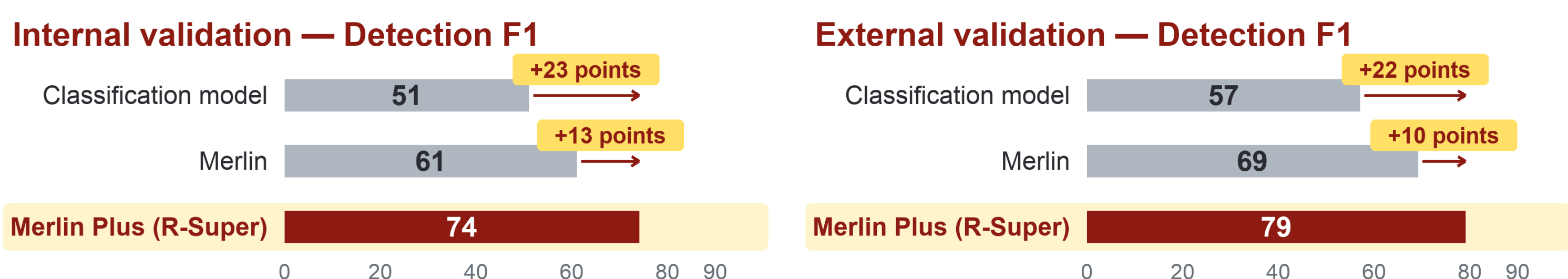
- ❖ We release anonymized patient IDs and scan dates
- ❖ Allows learning cancer evolution
- ❖ Large-scale longitudinal **CT scans, masks, and reports**



New State-of-the-art: Merlin Plus Models Substantially Surpass Previous Public AI in Tumor Detection & Segmentation



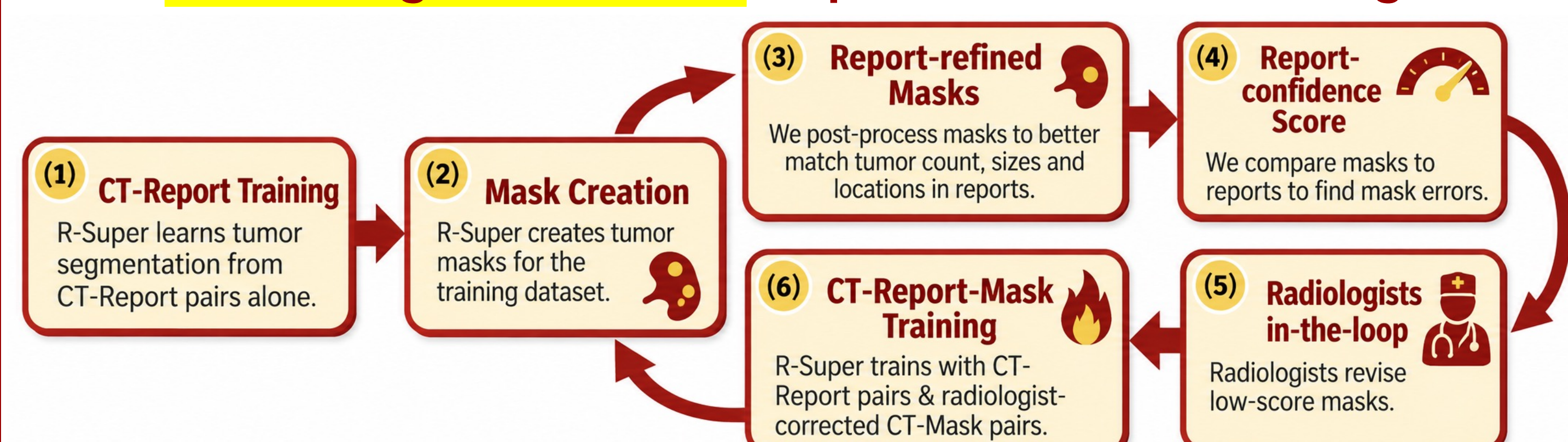
Masks Improve Cancer Detection: Merlin Plus Models Substantially Surpass Models Trained on Merlin (no Mask) in Tumor Detection



Full Results: Detection and Segmentation of 9 Tumor Types Internal & External Validation

internal validation (Merlin Plus)																							
model	supervision	spleen				bladder				gallbladder				adrenal				esophagus					
	report	mask	Se	Sp	F1	DSC	Se	Sp	F1	DSC	Se	Sp	F1	DSC	Se	Sp	F1	DSC	Se	Sp	F1	DSC	
public multi-tumor segmentation model																							
ULS [14]		x	34	78	44	3	42	66	35	12	23	92	33	3	2	100	4	0	17	99	25	0	
FLARE25 baseline [22]		x	12	100	21	4	12	100	21	0	13	100	23	1	17	100	28	6	0	100	0	0	
FLARE23 top1 [22]		x	20	99	32	3	2	100	5	0	21	100	34	13	13	100	24	5	0	100	0	0	
FLARE24 top1 [22]		x	19	100	31	2	5	100	9	0	19	100	32	9	20	100	33	11	0	100	0	0	
trained on Merlin classification																							
	x		58	58	58	-	46	80	46	-	72	38	47	-	57	67	63	-	55	83	44	-	
trained on Merlin Plus																							
MedFormer [13]		x	66	84	72	9	79	85	72	21	76	72	64	20	76	85	82	26	75	66	40	25	
R-Super [2]		x	86	77	82	25	86	91	82	29	77	93	80	43	83	96	89	41	80	71	46	33	
95% CI ($\pm$ half-w.)			6	8	5	9	11	5	9	10	11	5	8	13	6	3	4	15	25	6	25	17	
model	supervision		stomach				duodenum				uterus				prostate				average				
ULS [14]		x	47	59	41	4	28	93	37	2	49	68	51	13	60	79	50	27	34	82	35	7	
FLARE25 baseline [22]		x	30	97	44	2	11	96	19	0	50	87	59	21	32	87	36	8	20	96	28	5	
FLARE23 top1 [22]		x	26	100	42	1	6	97	10	0	5	100	10	3	3	100	6	2	11	100	18	3	
FLARE24 top1 [22]		x	30	98	45	3	11	99	20	0	26	99	40	18	6	99	12	2	15	100	25	5	
classification		x		61	52	48	-	56	68	44	-	67	68	63	-	52	80	46	-	58	66	51	-
MedFormer [13]		x	75	87	75	3	66	68	49	2	92	61	73	25	74	95	77	32	75	78	67	18	
R-Super [2]		x	77	90	78	16	78	70	57	9	73	87	76	47	84	92	79	51	80	85	74	33	
95% CI ($\pm$ half-w.)			12	4	9	7	14	8	12	8	10	6	7	14	14	4	12	18	4	2	4	4	
external validation																							
model	supervision		spleen				bladder				gallbladder				adrenal				esophagus				
public multi-tumor segmentation model																							
ULS [14]		x	66	51	69	4	62	72	63	9	47	82	56	0	15	96	26	0	43	78	52	11	
FLARE25 baseline [22]		x	20	96	32	12	7	99	13	4	3	98	7	6	1	100	2	1	6	98	12	2	
FLARE23 top1 [22]		x	28	97	43	9	1	99	1	0	3	98	6	3	6	100	11	1	0	100	0	0	
FLARE24 top1 [22]		x	22	95	36	13	1	100	1	0	3	99	7	5	1	100	3	1	7	100	14	4	
trained on Merlin classification																							
	x		59	51	64	-	68	40	56	-	59	42	54	-	55	60	63	-	78	60	72	-	
trained on Merlin Plus																							
MedFormer [13]		x	65	67	72	23	57	86	65	27	66	70	67	5	83	83	86	26	88	77	84	22	
R-Super [2]		x	74	78	80	53	75	79	75	32	71	72	71	19	88	92	92	50	84	82	83	32	
95% CI ($\pm$ half-w.)			4	6	3	8	7	6	5	8	7	6	5	16	3	4	2	8	6	6	4	9	
inter-reader			-	-	-	63	-	-	-	48	-	-	-	52	-	-	-	58	-	-	-	49	
model	supervision		stomach				duodenum				uterus				prostate				average				
ULS [14]		x	44	62	57	2	47	82	49	-	77	74	73	12	83	75	53	-	54	74	56	5	
FLARE25 baseline [22]		x	17	95	29	2	12	97	20	-	48	94	61	13	17	96	24	-	15	97	22	6	
FLARE23 top1 [22]		x	18	100	31	0	10	100	17	-	9	99	17	1	3	99	5	-	9	99	15	2	
FLARE24 top1 [22]		x	15	97	26	2	7	98	12	-	35	97	50	10	0	97	0	-	10	98	17	5	
classification		x		44	62	57	-	49	65	42	-	80	70	73	-	57	61	32	-	61	57	57	-
MedFormer [13]		x	59	65	70	2	57	85	59	-	83	77	78	22	89	95	83	-	72	78	74	18	
R-Super [2]		x	65	78	76	13	74	79	66	-	78	89	81	38	89	97	86	-	78	83	79	34	
95% CI ($\pm$ half-w.)			4	6	3	4	10	6	9	-	8	3	6	14	10	3	9	-	2	2	2	4	
inter-reader			-	-	-	46	-	-	-	47	-	-	-	58	-	-	-	54	-	-	-	53	

Accelerating Mask Creation: Report-Based Active Learning



- ❖ **New post-processing: auto refine masks to match tumor descriptions in reports!**
- ❖ Problematic masks are found by comparing masks and reports
- ❖ Radiologists revise refined masks: 3X – 5X faster than creating masks from scratch